

```
Last login: Fri Nov 28 16:39:40 on ttys004
nithin@Mac ~ %
nithin@Mac ~ %
nithin@Mac ~ % cd Desktop/Sem-5/OS/xv6-public/xv6-source
nithin@Mac xv6-source % qemu
zsh: command not found: qemu
nithin@Mac xv6-source % clear
```

SeaBIOS (version rel-1.17.0-0-gb52ca86e094d-prebuilt.qemu.org)

iPXE (<http://ipxe.org>) 00:03.0 CA00 PCI2.10 PnP PMM+1EFD1EC0+1EF31EC0 CA00

Booting from Hard Disk..xv6...

cpu0: starting 0

sb: size 1000 nblocks 916 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap star

=== XV6 User Authentication System ===

Default users:

admin/admin (full access)  
user1/pass1 (read only)  
user2/pass2 (read+write)

=====

Username: admin

Password: admin

Login successful! Welcome admin (UID: 1)

Starting shell...

admin\$ ls

ls

|      |              |                  |
|------|--------------|------------------|
| ---- | .            | uid:0 1 1 512    |
| ---- | ..           | uid:0 1 1 512    |
| ---- | README       | uid:0 2 2 2286   |
| ---- | cat          | uid:0 2 3 16868  |
| ---- | echo         | uid:0 2 4 15760  |
| ---- | grep         | uid:0 2 5 19728  |
| ---- | init         | uid:0 2 6 17564  |
| ---- | kill         | uid:0 2 7 15828  |
| ---- | ln           | uid:0 2 8 15700  |
| ---- | ls           | uid:0 2 9 18992  |
| ---- | mkdir        | uid:0 2 10 15824 |
| ---- | rm           | uid:0 2 11 15804 |
| ---- | sh           | uid:0 2 12 36292 |
| ---- | wc           | uid:0 2 13 17176 |
| ---- | login        | uid:0 2 14 16260 |
| ---- | logout       | uid:0 2 15 15692 |
| ---- | whoami       | uid:0 2 16 15936 |
| ---- | chmod        | uid:0 2 17 16120 |
| ---- | reboot       | uid:0 2 18 15580 |
| ---- | cpubound     | uid:0 2 19 16044 |
| ---- | iobound      | uid:0 2 20 16108 |
| ---- | mixed        | uid:0 2 21 16124 |
| ---- | mlfqtest     | uid:0 2 22 17688 |
| ---- | ps           | uid:0 2 23 16384 |
| rw-- | console      | uid:0 3 24 0     |
| rw-- | testfile.txt | uid:1 2 25 47    |

admin\$ mlfqtest

mlfqtest

=====

MLFQ SCHEDULER TEST

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This test spawns multiple processes:

- 2 CPU-bound processes (should be demoted)
- 1 I/O-bound process (should stay high priority)
- 1 Mixed workload process

Watch for [MLFQ] messages showing:

- Process demotion to lower queues
- Priority boosts
- Aging promotions

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[CPU-BOUND 5] Starting CPU-intensive task...

[CPU-BOUND 5] Iteration 0/100

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[CPU-BOUND 5] Iteration 10/100
[CPU-BOUND 5] Iteration 20/100
[CPU-BOUND 5] Iteration 30/100
[CPU-BOUND 5] Iteration 40/100
[CPU-BOUND 5] Iteration 50/100
[CPU-BOUND 5] Iteration 60/100
[CPU-BOUND 5] Iteration 70/100
[CPU-BOUND 5] Iteration 80/100
[CPU-BOUND 5] Iteration 90/100
[CPU-BOUND 5] Completed!
[IO-BOUND 6] Starting I/O-intensive task...
[IO-BOUND 6] I/O operation 1/50 completed
[IO-BOUND 6] I/O operation 2/50 completed
[CPU-BOUND 7] Starting CPU-intensive task...
[CPU-BOUND 7] Iteration 0/100
[CPU-BOUND 7] Iteration 10/100
[CPU-BOUND 7] Iteration 20/100
[IO-BOUND 6] I/O operation 3/50 completed
[CPU-BOUND 7] Iteration 30/100
[CPU-BOUND 7] Iteration 40/100
[CPU-BOUND 7] Iteration 50/100
[CPU-BOUND 7] Iteration 60/100
[CPU-BOUND 7] Iteration 70/100
[CPU-BOUND 7] Iteration 80/100
[CPU-BOUND 7] Iteration 90/100
[CPU-BOUND 7] Completed!
[IO-BOUND 6] I/O operation 4/50 completed

[TEST] All processes spawned. Waiting for completion...

[MIXED 8] Starting mixed workload...
[MIXED 8] CPU phase 1
[MIXED 8] I/O phase 1
[IO-BOUND 6] I/O operation 5/50 completed
[IO-BOUND 6] I/O operation 6/50 completed
[MIXED 8] CPU phase 2
[MIXED 8] I/O phase 2
[IO-BOUND 6] I/O operation 7/50 completed
[IO-BOUND 6] I/O operation 8/50 completed
[MIXED 8] CPU phase 3
[MIXED 8] I/O phase 3
[IO-BOUND 6] I/O operation 9/50 completed
[IO-BOUND 6] I/O operation 10/50 completed
[MIXED 8] CPU phase 4
[MIXED 8] I/O phase 4
[IO-BOUND 6] I/O operation 11/50 completed
[IO-BOUND 6] I/O operation 12/50 completed
[MIXED 8] CPU phase 5
[MIXED 8] I/O phase 5
[IO-BOUND 6] I/O operation 13/50 completed
[IO-BOUND 6] I/O operation 14/50 completed
[MIXED 8] CPU phase 6
[MIXED 8] I/O phase 6
[IO-BOUND 6] I/O operation 15/50 completed
[IO-BOUND 6] I/O operation 16/50 completed
[MIXED 8] CPU phase 7
[MIXED 8] I/O phase 7
[IO-BOUND 6] I/O operation 17/50 completed
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[IO-BOUND 6] I/O operation 18/50 completed  
[MIXED 8] CPU phase 8  
[MIXED 8] I/O phase 8  
[IO-BOUND 6] I/O operation 19/50 completed  
[IO-BOUND 6] I/O operation 20/50 completed  
[MIXED 8] CPU phase 9  
[MIXED 8] I/O phase 9  
[IO-BOUND 6] I/O operation 21/50 completed  
[IO-BOUND 6] I/O operation 22/50 completed  
[MIXED 8] CPU phase 10  
[MIXED 8] I/O phase 10  
[IO-BOUND 6] I/O operation 23/50 completed  
[MIXED 8] CPU phase 11  
[MIXED 8] I/O phase 11  
[IO-BOUND 6] I/O operation 24/50 completed  
[IO-BOUND 6] I/O operation 25/50 completed  
[MIXED 8] CPU phase 12  
[MIXED 8] I/O phase 12  
[IO-BOUND 6] I/O operation 26/50 completed  
[IO-BOUND 6] I/O operation 27/50 completed  
[MIXED 8] CPU phase 13  
[MIXED 8] I/O phase 13  
[IO-BOUND 6] I/O operation 28/50 completed  
[IO-BOUND 6] I/O operation 29/50 completed  
[MIXED 8] CPU phase 14  
[MIXED 8] I/O phase 14  
[IO-BOUND 6] I/O operation 30/50 completed  
[IO-BOUND 6] I/O operation 31/50 completed  
[MIXED 8] CPU phase 15  
[MIXED 8] I/O phase 15  
[IO-BOUND 6] I/O operation 32/50 completed  
[IO-BOUND 6] I/O operation 33/50 completed  
[MIXED 8] CPU phase 16  
[MIXED 8] I/O phase 16  
[IO-BOUND 6] I/O operation 34/50 completed  
[IO-BOUND 6] I/O operation 35/50 completed  
[MIXED 8] CPU phase 17  
[MIXED 8] I/O phase 17  
[IO-BOUND 6] I/O operation 36/50 completed  
[IO-BOUND 6] I/O operation 37/50 completed  
[MIXED 8] CPU phase 18  
[MIXED 8] I/O phase 18  
[IO-BOUND 6] I/O operation 38/50 completed  
[IO-BOUND 6] I/O operation 39/50 completed  
[MIXED 8] CPU phase 19  
[MIXED 8] I/O phase 19  
[IO-BOUND 6] I/O operation 40/50 completed  
[IO-BOUND 6] I/O operation 41/50 completed  
[MIXED 8] CPU phase 20  
[MIXED 8] I/O phase 20  
[IO-BOUND 6] I/O operation 42/50 completed  
[IO-BOUND 6] I/O operation 43/50 completed  
[MIXED 8] Completed!  
[IO-BOUND 6] I/O operation 44/50 completed  
[IO-BOUND 6] I/O operation 45/50 completed  
[IO-BOUND 6] I/O operation 46/50 completed  
[IO-BOUND 6] I/O operation 47/50 completed  
[IO-BOUND 6] I/O operation 48/50 completed

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[IO-BOUND 6] I/O operation 49/50 completed
[IO-BOUND 6] I/O operation 50/50 completed
[IO-BOUND 6] Completed!
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=====
MLFQ TEST COMPLETED
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=====
Check the output above for:
1. CPU-bound processes being demoted
2. I/O-bound process staying responsive
3. Priority boosts occurring
4. Fair CPU distribution
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admin$
```